

7 Fig. 7.1 shows some of the energy levels of an helium atom.



Fig. 7.1

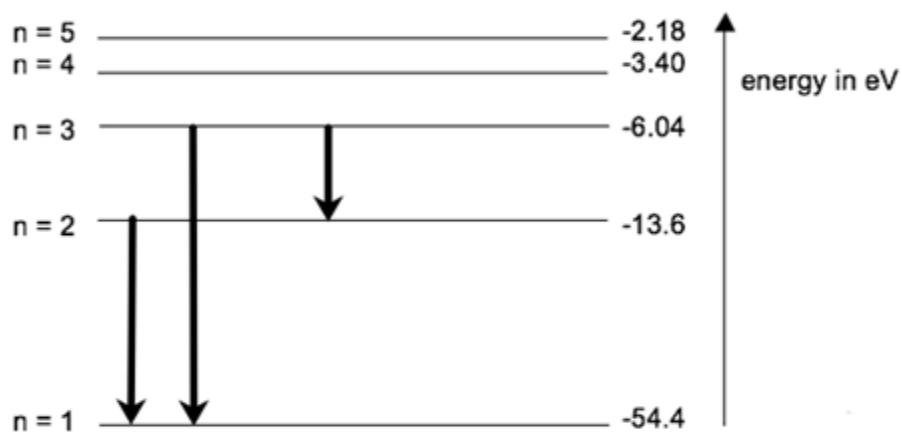
An electron with kinetic energy of 50.0 eV collides with a helium atom in its ground state and the helium atom is excited.

- (a) (i) In Fig. 7.1, use arrows to show the possible energy transitions when the excited helium atom de-excites.

[2]

Solution:

Highest possible level that the electron can excite to is -6.04 eV, $n = 3$



2 marks - all 3 transitions
1 mark – any 2 transitions
0 mark – 0 or 1 transition

		(ii) Calculate the shortest wavelength of the radiation that is emitted from the transitions in (a)(i).
		Solution: Loss of electron excitation energy ΔE = Energy of emitted photon E_{photon} $\Delta E = \frac{hc}{\lambda}$ Largest ΔE (i.e. from $n=3$ to $n=1$) produces photon of shortest wavelength λ. $[-6.04 - (-54.4)](1.6 \times 10^{-19}) = \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{\lambda}$ $\lambda = 2.57 \times 10^{-8} \text{ m}$
		shortest wavelength = m [2]
		(b) When a beam of white light is passed through a cold helium gas, an absorption spectrum of coloured background with dark lines is observed. Use Fig. 7.1 to explain quantitatively why one of the dark lines correspond to a wavelength of 471 nm.
		
		 [3]
		Solution: Photon of wavelength 471 nm has energy of $\frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{471 \times 10^{-9}} = 2.64 \text{ eV}$ Electrons absorb photons of energy 2.64 eV to make a transition from $n = 3$ to $n = 4$. 1 On de-excitation, photons of energy 2.64 eV, i.e. wavelength 471 nm are emitted. 1 But photons emitted are in all directions, not just along the initial direction, hence dark line is observed as compared to other regions whose wavelengths are not absorbed. 1

Section B

Answer **one** question from this Section in the spaces provided.

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| 8 | (a) | Two ions S and T, each of negative charge $-q$, are held stationary at a distance of 2.0 cm from |
|---|-----|---|